Chemistry GCE O Level 2019 - Paper 1

Duration: 1h30 | Cameroon GCE Board

Multiple choice question paper, 0515 Chemistry 1, June 2019 (Cameroon GCE Board, Ordinary Level). One and a half hours. Answer ALL 50 questions. Each question has four suggested answers A, B, C and D; mark only one answer. Calculators are allowed.

Useful data. Relative atomic masses: H = 1.0, C = 12.0, O = 16.0, Cl = 35.5, N = 14.0, Na = 23.0, Cu = 64.0, Zn = 65.0. 1 Faraday = 96000 coulombs. Molar volume of any gas at r.t.p. = 24000 cm^3. Avogadro number = 6.02×10^{23}. Specific heat capacity of water = $4.2\ \text{J}\ \text{g}^{-1}\ ^{\circ}\text{C}^{-1}$.

Multiple choice questions - choose the one correct answer, A, B, C or D.

1. Choose a metal commonly used as overhead electric cables.

a) Zinc
b) Aluminium
c) Iron
d) Copper

2. Select a method that can be used to separate a mixture of water and petrol.

a) Evaporation to dryness
b) Separating funnel
c) Sublimation
d) Simple distillation

3. Determine the atomic number of an atom with electronic configuration 2,8,8,2.

a) 10
b) 4
c) 2
d) 20

4. The thermal decomposition of calcium nitrate will produce;

a) $\text{Ca}(\text{NO}_2)_2$ and O_2
b) NO_2 and O_2
c) N_2 and O_2
d) N_2O and O_2

5. The transition metal used as a catalyst in the manufacture of ammonia is

a) Vanadium
b) Copper
c) Iron
d) Platinum

6. The element X exists naturally as two isotopes, ^{80}X and ^{82}X in the ratio 1 : 1. The relative atomic mass of X will be

a) 81
b) 80.5
c) 82
d) 80

7. *Questions 7 - 9 concern the following table.*

Element	M	N	O	P	Q
Proton	12	16	17	18	20
Electron	12	17	17	18	20

M, N, O, P and Q are not the usual symbols of the elements. Each element can be selected once, more than once or not at all. Select from the table:

1. An element that will not react easily with other elements.

a) M
b) N
c) O
d) P

8. *Questions 7 - 9 concern the following table.*

Element	M	N	O	P	Q
Proton	12	16	17	18	20
Electron	12	17	17	18	20

M, N, O, P and Q are not the usual symbols of the elements. Each element can be selected once, more than once or not at all. Select from the table:

1. Elements that belong to the same group of the Period Table

a) M and N
b) M and Q
c) O and Q
d) N and O

9. *Questions 7 - 9 concern the following table.*

Element	M	N	O	P	Q
Proton	12	16	17	18	20

Element	M	N	O	P	Q
Electron	12	17	17	18	20

M, N, O, P and Q are not the usual symbols of the elements. Each element can be selected once, more than once or not at all. Select from the table:

1. Determine the formula of the compound formed between M and N

a) $\mathrm{M_2N_2}$
b) MN
c) $\mathrm{MN_2}$
d) $\mathrm{M_2N}$

10. Give the formula of the fifth member of the homologous series with general formula $\mathrm{C_nH_{2n+2}}$

a) $\mathrm{C_5H_{12}}$
b) $\mathrm{C_5H_{10}}$
c) $\mathrm{C_5H_{11}}$
d) $\mathrm{C_5H_8}$

11. Name two air pollutants which are responsible for acid rain.

a) $\mathrm{N_2}$ and $\mathrm{SO_2}$
b) $\mathrm{NO_2}$ and $\mathrm{CO_2}$
c) $\mathrm{CO_2}$ and $\mathrm{SO_2}$
d) $\mathrm{SO_2}$ and $\mathrm{NO_2}$

12. Which of the following substances will burn with a smoky and luminous flame?

a) Propane
b) Ethene
c) Hydrogen
d) Butane

13. Calculate the amount of substance in moles in $250\ \mathrm{cm^3}$ of $0.1\ \mathrm{M\ HCl(aq)}$.

a) 0.05
b) 0.25
c) 0.025
d) 0.5

14. When excess carbon dioxide is passed through lime water, the milky colour disappears due to the formation of

a) $\mathrm{Ca(HCO_3)_2}$
b) $\mathrm{CaCO_3}$
c) CaO
d) $\mathrm{Ca(OH)_2}$

15. What is the colour imparted to a Bunsen flame by a copper (II) salt?

a) Bluish green
b) Apple green

c) Golden yellow
d) Brick red

16. The apparatus used in preparing a standard solution is a

a) Burette
b) Measuring cylinder
c) Pipette
d) Volumetric flask

17. During an oxidation reaction, an oxidizing agent

a) Gains oxygen
b) Loses oxygen
c) Loses hydrogen
d) Loses electrons

18. Calculate the quantity of electricity in coulombs if a current of $0.5\ \mathrm{A}$ is passed through an electrolyte for 15 minutes.

a) 450
b) 45
c) 7.5
d) 27000

19. The number of replaceable hydrogen atoms in a molecule of an acid is called its

a) Acidity
b) Alkalinity
c) Basicity
d) pH value

20. The substance left in the glass tube when excess dry hydrogen is passed over heated copper (II) oxide is:

a) Copper(I) oxide
b) Water
c) Oxygen
d) Metallic copper

21. The function of limestone in the extraction of iron is to remove

a) Excess air
b) Earthy impurities
c) Unburnt coke
d) Molten iron

22. *Instructions: each question is followed by four responses numbered 1 - 4. One or more of these responses is (are) correct. Decide which response(s) is (are) correct, then choose:* A) If only 1, 2 and 3 are correct; B) If only 1 and 3 are correct; C) If only 2 and 4 are correct; D) If only 4 is correct.

1. The factors that influence the preferential discharge of ions during electrolysis are

1 Position of ion on the electrochemical series.

2 Nature of the electrode.

3 Concentration of the ion.

4 The charge on the ion.

a) 1, 2 and 3 only
b) 1 and 3 only
c) 2 and 4 only
d) 4 only

23. *Instructions: each question is followed by four responses numbered 1 - 4. One or more of these responses is (are) correct. Decide which response(s) is (are) correct, then choose:* A) If only 1, 2 and 3 are correct; B) If only 1 and 3 are correct; C) If only 2 and 4 are correct; D) If only 4 is correct.

1. A catalyst is a substance which

1 Increases the temperature of a reaction

2 Increases the rate of a chemical reaction

3 Increases the yield of a chemical reaction

4 Remains chemically unchanged at the end of a reaction

a) 1, 2 and 3 only
b) 1 and 3 only
c) 2 and 4 only
d) 4 only

24. The components of crude oil can be separated by

a) Reforming
b) Cracking
c) Simple distillation
d) Fractional distillation

25. The empirical formula of an organic compound is CH. If the relative molecular mass of the compound is 78, then the molecular formula is:

a) C_2H_2
b) C_6H_6
c) C_4H_4
d) C_6H_{12}

26. Carbon dioxide is prepared in the laboratory by the action of dilute hydrochloric acid on marble chips. Dilute sulphuric acid is not suitable for this process because;

a) An insoluble coat of calcium sulphate will be formed causing the reaction to stop.
b) The reaction will be too fast.
c) The reaction will be too slow.
d) Marble chips do not react with dilute sulphuric acid.

27. Select a carbonate that will NOT decompose when strongly heated in an open crucible.

a) $CaCO_3$
b) $CuCO_3$
c) Na_2CO_3
d) $MgCO_3$

28. *Questions 28 - 30 concern the following substances:* A) Ammonia gas, B) Lemon juice, C) Water, D) Ethanol. Each letter can be used once, more than once or not at all. Select from A to D a substance

1. Which produces steamy fumes with PCl_5.

Options as printed: A) Ammonia gas B) Lemon juice C) Water D) Ethanol

29. *Questions 28 - 30 concern the following substances:* A) Ammonia gas, B) Lemon juice, C) Water, D) Ethanol. Each letter can be used once, more than once or not at all. Select from A to D a substance

1. That will turn moist red litmus paper blue.

a) Ammonia gas
b) Lemon juice
c) Water
d) Ethanol

30. *Questions 28 - 30 concern the following substances:* A) Ammonia gas, B) Lemon juice, C) Water, D) Ethanol. Each letter can be used once, more than once or not at all. Select from A to D a substance

1. That will turn blue litmus paper red.

a) Ammonia gas
b) Lemon juice
c) Water
d) Ethanol

31. Determine the percentage by mass of nitrogen in NH_4NO_3. (RMM of $NH_4NO_3 = 80$)

a) 17.5
b) 70
c) 35
d) 22.5

32. Which of these halogens is the most reactive?

a) Chlorine
b) Bromine
c) Iodine
d) Fluorine

33. Which of the following occurs during the electrolysis of aqueous copper (II) sulphate using copper electrodes?

a) The cathode increases in size
b) The anode increase in size
c) The colour of the solution fades
d) Oxygen is discharged at the anode

34. The compound most likely to be present in a bag of fertilizer is:

a) $Ca(NO_3)_2$
b) $(NH_4)_3PO_4$
c) NH_4Cl
d) $NaCl$

35. *Instructions: each question consists of a statement in the left-hand column followed by another in the right-hand column. Decide whether each of the statements is true or false, then choose:* A) If both statements are TRUE and the second statement is the CORRECT EXPLANATION of the first; B) If both statements are TRUE but the second statement is NOT the correct explanation of the first; C) If the first statement is true but the second statement is FALSE; D) If the first statement is FALSE but the second statement is TRUE.

1. **First statement:** Elements of Group VII show similar chemical properties.

Second statement: Group VII elements have same number of shells.

a) Both true, second explains first
b) Both true, second does not explain first
c) First true, second false
d) First false, second true

36. *Instructions: each question consists of a statement in the left-hand column followed by another in the right-hand column. Decide whether each of the statements is true or false, then choose:* A) If both statements are TRUE and the second statement is the CORRECT EXPLANATION of the first; B) If both statements are TRUE but the second statement is NOT the correct explanation of the first; C) If the first statement is true but the second statement is FALSE; D) If the first statement is FALSE but the second statement is TRUE.

1. **First statement:** The presence of Cl^-, Br^- and I^- ions in solution can be tested using $AgNO_3$ solution.

Second statement: All three ions form coloured precipitates with $AgNO_3$ solution.

Options as printed: A) Both true, second explains first B) Both true, second does not explain first C) First true, second false D) First false, second true

37. *Instructions: each question consists of a statement in the left-hand column followed by another in the right-hand column. Decide whether each of the statements is true or false, then choose:* A) If both statements are TRUE and the second statement is the CORRECT EXPLANATION of the first; B) If both statements are TRUE but the second statement is NOT the correct explanation of the first; C) If the first statement is true but the second statement is FALSE; D) If the first statement is FALSE but the second statement is TRUE.

1. **First statement:** Copper will displace magnesium from a solution of a magnesium salt.

Second statement: Copper is below magnesium in the electrochemical series.

a) Both true, second explains first
b) Both true, second does not explain first
c) First true, second false
d) First false, second true

38. Palm wine goes sour when exposed to air for some days because;

a) Glucose is converted to gluconic acid
b) Ethanol is converted to Ethanoic acid
c) Glucose is converted to ethanol
d) Palm wine is converted to palmitic acid

39. Calculate the mass of zinc that will produce $16\ \text{g}$ of copper according to the equation:

$$\text{Zn(s)} + \text{CuSO}_4\text{(aq)} \rightarrow \text{ZnSO}_4\text{(aq)} + \text{Cu(s)}$$

a) 1.625
b) 15.75
c) 16.25
d) 1.58

40. Which of the following is a basic oxide?

a) Al_2O_3
b) SO_2
c) Na_2O
d) CO

41. What is the mass of 0.1 mole of Na_2CO_3? (RMM of $\text{Na}_2\text{CO}_3 = 106$)

a) 106 g
b) 10.0 g
c) 1.06 g
d) 10.6 g

42. Plastics pollute the environment because they are;

a) Insoluble in water
b) Non-biodegradable
c) Biodegradable
d) Inflammable

43. The different components of universal indicator solution can best be separated by

a) Fractional distillation
b) Filtration
c) Chromatography
d) Simple distillation

44. Consider the equation

$$\text{NaOH(aq)} + \text{HCl(aq)} \rightarrow \text{NaCl(aq)} + \text{H}_2\text{O(l)}\ \Delta H = X\ \text{kJ}$$

What does X represent?

a) Heat of neutralization
b) Activation energy
c) Heat of formation
d) Heat of solution

45. $0.1\ \text{M HCl}$ reacts with powdered Na_2CO_3 according to the equation

$$2\text{HCl}(\text{aq}) + \text{Na}_2\text{CO}_3(\text{s}) \rightarrow 2\text{NaCl}(\text{aq}) + \text{H}_2\text{O}(\text{l}) + \text{CO}_2(\text{g}).$$

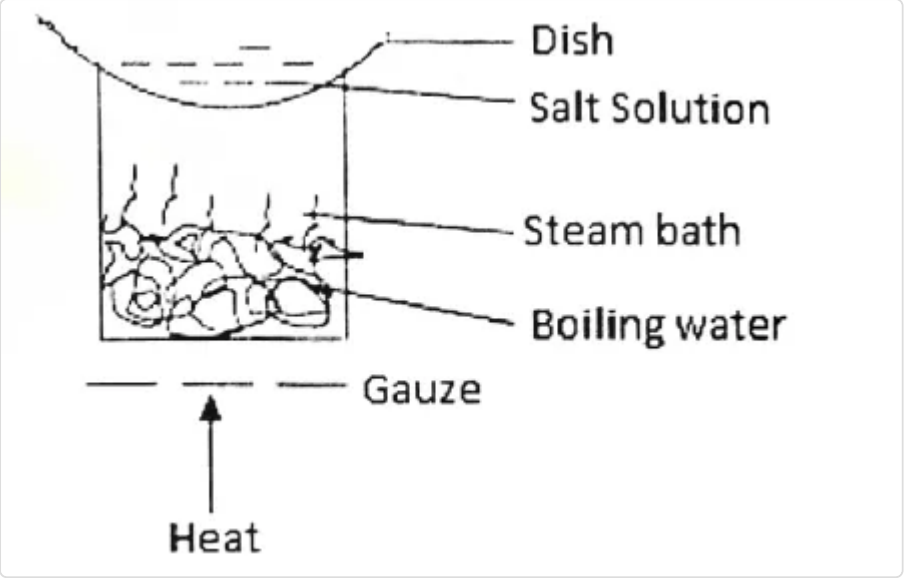
How can the reaction be made to go faster?

a) Using 0.5 M HCl
b) Using lumps of Na_2CO_3
c) Adding water to the reaction mixture
d) Using 0.01 M HCl

46. How many particles of NaOH are contained in 0.1 mole of NaOH?

a) 6.02×10^{23}
b) 6.02×10^{24}
c) 6.02×10^{25}
d) 6.02×10^{22}

47. *Questions 47 and 48 are based on the diagram below.*

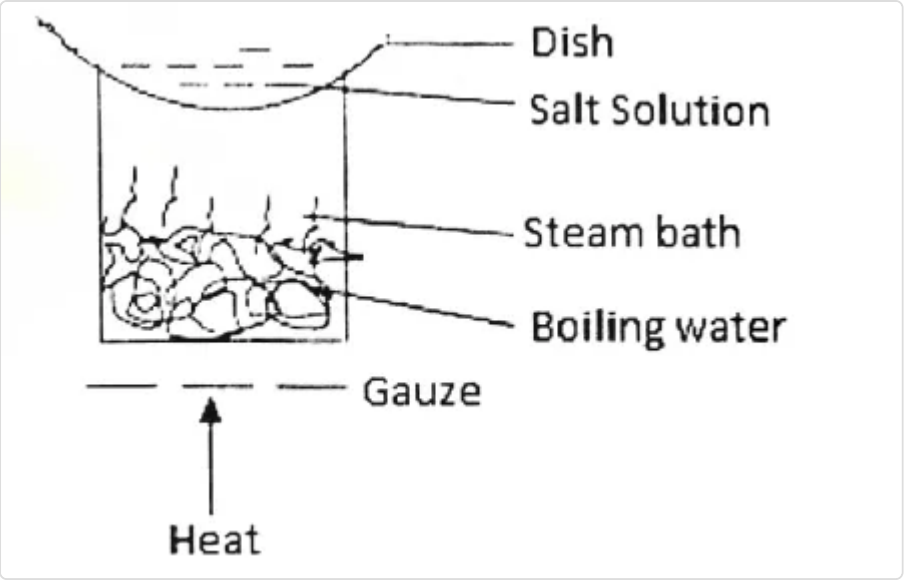


The diagram shows an evaporating dish holding salt solution, placed over a beaker of boiling water (steam bath) on a gauze, heated from below.

1. What method of separation is illustrated in the diagram?

Options as printed: A) Crystallization B) Sublimation C) Evaporation to dryness D) Filtration

48. *Question 48 is based on the diagram of question 47 (a dish of salt solution heated over a steam bath).*



1. An example of a mixture that can be separated by this method is

a) Kerosene and water
b) Ethanol and water
c) NH_4Cl and $NaCl$
d) $NaCl$ and H_2O

49. Select the process that involves a physical change.

a) Heating candle wax
b) Electrolyzing dil. NaCl
c) Burning a piece of Na
d) Corrosion

50. Isotopes of a given element have the same

a) Mass number
b) Number of neutrons
c) Physical properties
d) Atomic number

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